

derivation (version s_0)

LLM agent derives K through deterministic tool calls (trace)

(1) record evidence: anchor calls by symbol, content or key

store K, anchored queries, state e_0

every later read (version s_t)

(2) replay anchored queries (no LLM), giving state e_t

e_t **equal to anchor state?**

yes

reuse K
(no LLM tokens)

no

(3) memo hit on (anchor, e_t)? else build focused delta; split hierarchically if large

LLM judge:
still_valid |
changed | unsure

↑still_valid

changed /
unsure

(4) re-derive with the agent at s_t ; new K, new anchor

(5) memoise verdict per (anchor, state)

LLM cost grows with the number of distinct evidence states, not with reads or commits